

Survey of Reasoning-Based Autonomous Driving in Mixed Traffic: An Offline–Online Two-Loop Perspective

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Manuscript under review at IEEE Transactions on Intelligent Transportation Systems (T-ITS), 2026

Abstract—Autonomous vehicles operating in mixed traffic networks frequently encounter unpredictable, interactive scenarios characterized by complex human-machine interactions and partial observability. While traditional motion planning approaches have made significant strides, this paper explores the emerging paradigm of reasoning-based autonomous driving and its impact on broader transportation operations. By integrating vision-language-action models, this paradigm shifts from purely reactive behaviors to explicit deliberation through intermediate representations, such as rationales and structured plans. To organize this rapidly evolving field, we adopt a unified two-loop offline and online framework. The offline loop focuses on acquiring generalized driving priors, including foundational policies, world models, and evaluators, through supervised learning and reinforcement learning techniques. The online loop then refines these learned priors at decision time through mechanisms such as test-time adaptation, sampling-based predictive control, and lookahead tree search. We synthesize how reasoning tokens and world models bridge these two loops to enable superior decision-making. Finally, we outline future research directions and open questions for achieving safer, more robust autonomy in mixed traffic.

Index Terms—Reasoning, Mixed traffic, Artificial Intelligence, Autonomous driving

I. INTRODUCTION

Autonomous vehicles (AVs) deployed in mixed traffic struggle with long-tail, interactive scenarios marked by heterogeneous agents and partial observability. Large Language Models (LLMs) and Vision-Language models (VLMs), as well as Vision-Language-Action models (VLAs), offer a promising solution by incorporating explicit reasoning, which is represented as intermediate tokens (e.g., rationales, structured plans) in language models, to enable deliberation. We call this paradigm as reasoning-based autonomous driving (AD): shifting from

purely reactive policies to those utilizing intermediate representations and computation for superior decision-making.

To organize this emerging literature, we adopt Bertsekas’s two-loop offline-online framework [1]. The offline loop acquires generalized priors, such as policies, value approximations, and world models, while the online loop refines these decisions at inference time via bounded search or adaptation. Crucially, VLM/VLA-based reasoning expands this paradigm: online improvement often occurs internally via intermediate-token deliberation rather than solely relying on external planners.

Existing surveys have provided valuable overviews of AD and classical motion planning or decision-making. For example, recent End-to-End (E2E) driving surveys emphasize architectural paradigms and unified perspectives on sensor-to-action learning [2]. Complementarily, motion planning surveys review optimization-, sampling-, and learning-based planning and control techniques [3]–[5]. For mixed-autonomy control, [6] provides a survey spanning physics-based control, game-theoretic modeling, and AI-guided policy learning in mixed traffic. In contrast, our focus is on *reasoning as an offline–online coupling mechanism*: how intermediate tokens, evaluators (including learned reward models), and world models enable deliberation and verification for mixed-traffic interaction, and how these components connect modern LLM/VLM post-training pipelines with planning and control.

- **A systematic analysis of reasoning-based autonomy:** We examine how reasoning bridges the gap between perception and control through deliberation.
- **A unified offline–online taxonomy:** We formalize the AD stack into a cohesive two-loop framework, categorizing the offline acquisition of driving priors and their online refinement.
- **Synthesis of future directions:** We outline promising future directions and insights for reasoning-based autonomous driving in mixed traffic.

The structure of this survey maps directly to the two-loop view: covering offline learning primitives, online improvement mechanisms, and exploring how reasoning tokens and world models bridge these loops to tackle mixed-traffic deployment.

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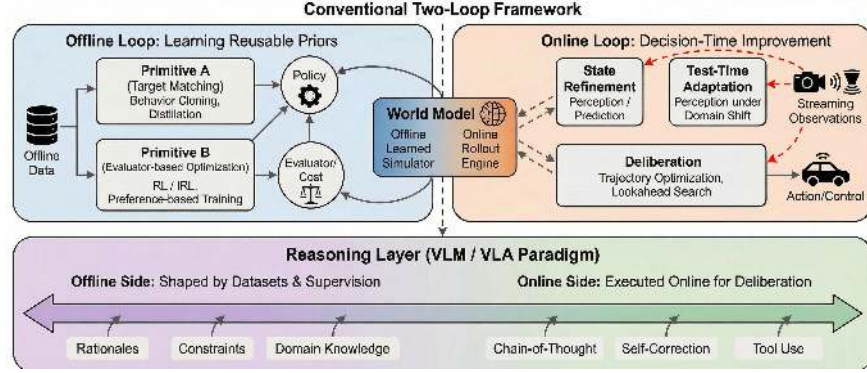


Fig. 1. Survey roadmap under an offline-online two-loop view. The offline loop learns reusable priors (policies, world models, and evaluators). The online loop improves decisions at deployment (state refinement, adaptation, and deliberation). The Reasoning Layer (VLM/VLA paradigm) bridges these two loops by utilizing intermediate tokens such as rationales, constraints, and chain-of-thought to enable deliberation.

II. OFFLINE

Following the two-loop framework [1], the *offline* loop acquires generalized priors from static data or simulation. In mixed traffic, these priors are supposed to encode both low-level kinematic feasibility and high-level interaction patterns, enabling the subsequent *online* loop to refine decisions under real-time constraints.

We categorize offline policy learning into two foundational primitives based on their mathematical objectives:

- 1) **Primitive A: Target Matching** (Supervised Learning/Distillation). The policy is trained to match an explicit target conditional distribution over actions or trajectories.
- 2) **Primitive B: Evaluator-Guided Optimization** (RL/IRL). The policy is optimized to maximize scores from an evaluator that assesses candidate behaviors.

While both primitives aim to produce competent driving policies, they diverge fundamentally in their optimization targets. In practice, recent systems frequently pipeline both (e.g., Supervised Fine-tuning (SFT) followed by RL fine-tuning). Our goal is not to force every paper into a single box, but to provide a clean conceptual decomposition. And this decomposition will guide the remainder of the offline section.

A. Primitive A: Target Matching

Primitive A frames offline learning as conditional distribution matching. Given context $x \in \mathcal{X}$ and output $y \in \mathcal{Y}$, the policy $\pi_\theta(y | x)$ minimizes divergence from a target $\mu(y | x)$ under an input distribution $q(x)$:

$$\min_{\theta} \mathbb{E}_{x \sim q(x)} [\text{KL}(\mu(\cdot | x) \parallel \pi_\theta(\cdot | x))]. \quad (1)$$

This formulation yields a 2x2 design space (Figure 2) governed by two axes: (i) **Input distribution** $q(x)$: static datasets (q_{data}) versus dynamic, on-policy rollouts (q_{π_θ}).

Legend: LLM Language domain ($x \rightarrow y$), AD Driving domain ($x \rightarrow y$), Eq. Learning objective.

	Fixed-data inputs $x \sim q_{\text{data}}(x)$	On-policy inputs $x \sim q_{\pi_\theta}(x)$
Hard target (single label)	A.1 SFT / BC Eq. (2) dataset prompt $\rightarrow$ gold response logged state $\rightarrow$ expert trajectory	A.4 Dagger-style Eq. (6) sampled prefix $\rightarrow$ oracle correction visited state $\rightarrow$ expert takeover
Soft target (teacher distribution)	A.2 Off-policy KD Eq. (3) dataset prompt $\rightarrow$ teacher logits logged state $\rightarrow$ teacher act dist.	A.3 On-policy distillation Eq. (4) sampled prefix $\rightarrow$ teacher logits visited state $\rightarrow$ teacher act dist.

Fig. 2. Primitive A (offline target matching) by two knobs: where inputs x come from (fixed dataset vs. on-policy rollouts) and what supervision provides for each x (hard label vs. soft teacher distribution).

(ii) **Target conditional** $\mu(\cdot | x)$: hard point-mass labels ($\delta_{y^*(x)}$) versus soft teacher distributions ($p_T(\cdot | x)$).

a) **A.1 SFT / Behavior Cloning (Hard Targets, Fixed Data)**: This quadrant represents standard maximum-likelihood estimation, where $q = q_{\text{data}}$ and $\mu(\cdot | x) = \delta_{y^*(x)}$. In driving, this maps a logged state to an expert trajectory:

$$\min_{\theta} \mathbb{E}_{(x, y^*) \sim \mathcal{D}} [-\log \pi_\theta(y^* | x)]. \quad (2)$$

In LLM settings, SFT is maximum likelihood on (instruction, response) pairs; the “expert” is often a mixture of sources (human-written, model-generated, curated corpora, tool-augmented traces, etc.), so both q_{data} and y^* can be heterogeneous.

Behavior cloning (BC) uses the same objective (2) but is typically discussed in imitation-learning contexts.

SFT is foundational for LLM alignment via instruction tuning (InstructGPT, Self-Instruct [7], [8]), while BC is also the standard baseline for AD. Recent works apply BC across diverse architectures, from transformer-based sensor fusion (TransFuser [9]) and safety-constrained feature extraction (InterFuser [10]) to Learning from

All Vehicles (LAV) [11]. With the rise of VLMs, SFT is increasingly used to bind language instructions to driving actions (LMDrive [12]), inject QA-based spatial reasoning (DriveLM, DriveVLM, VLAD [13]–[15]), and refine sensorimotor agents via LLM-generated feedback or chain-of-thought data (FeD, AutoDrive-R² [16], [17]).

b) A.2 Offline Knowledge Distillation (Soft Targets, Fixed Data): Here, the model matches a teacher distribution over a fixed dataset ($q = q_{\text{data}}$ and $\mu(\cdot | x) = p_T(\cdot | x)$):

$$\min_{\theta} \mathbb{E}_{x \sim q_{\text{data}}} [\text{KL}(p_T(\cdot | x) \| \pi_{\theta}(\cdot | x))]. \quad (3)$$

Distillation via soft targets was popularized by [18], and was originally designed for model compression [18], [19]. Autonomous driving systems leverage offline Knowledge Distillation (KD) to transfer driving knowledge to deployable models. Applications include distilling heavy (and diversified) end-to-end planners into lighter variants (PlanKD, DistillDrive [20], [21]), compressing multi-modal LLM reasoning into purely vision-based sensorimotor students (DiMA, CaT [22], [23]), and mitigating catastrophic forgetting in continual learning (VLM-Assisted [24]).

c) A.3 On-Policy Distillation (Soft Targets, Dynamic Inputs): To address covariate shift, i.e., the distribution of input data shifts between training and deploying environments, this paradigm matches a teacher on the student's own rollouts ($q = q_{\pi_{\theta}}$):

$$\min_{\theta} \mathbb{E}_{x \sim q_{\pi_{\theta}}} [\text{KL}(p_T(\cdot | x, c) \| \pi_{\theta}(\cdot | x))]. \quad (4)$$

In LLMs, this is an emerging direction. Generating training prefixes on-policy mitigates exposure bias. Typical methods utilize teacher feedback on student generated sequences (GKD [25]), or a single model as both teacher and student by conditioning on different contexts (i.e., the additional context c for teacher) (OPSD [26]), or leverage different divergence (i.e., reverse KL) to encourage mode-seeking behavior when distill (OPCD [27]).

In driving, explicit “on-policy distillation” is less standard; the most common and closely related practice is DAgger-style data aggregation.

d) A.4 DAgger (Hard Targets, Dynamic Inputs via Aggregation): DAgger [28] iteratively updates the training distribution toward states visited by the learner while keeping hard labels from an expert:

$$\mathcal{D}_{k+1} \leftarrow \mathcal{D}_k \cup \{(x, y^*(x)) : x \sim q_{\pi_{\theta_k}}\}, \quad (5)$$

$$\theta_{k+1} \leftarrow \arg \min_{\theta} \mathbb{E}_{(x, y^*) \sim \mathcal{D}_{k+1}} [-\log \pi_{\theta}(y^* | x)]. \quad (6)$$

In the AD context, recent pipelines studies DAgger-style techniques in CARLA, emphasizing critical-state sampling and replay-buffer design under high-uncertainty

regions [29]. TakeAD [30] combines iterative DAgger (to handle disengagement/takeover states) with a subsequent stage for preference alignment.

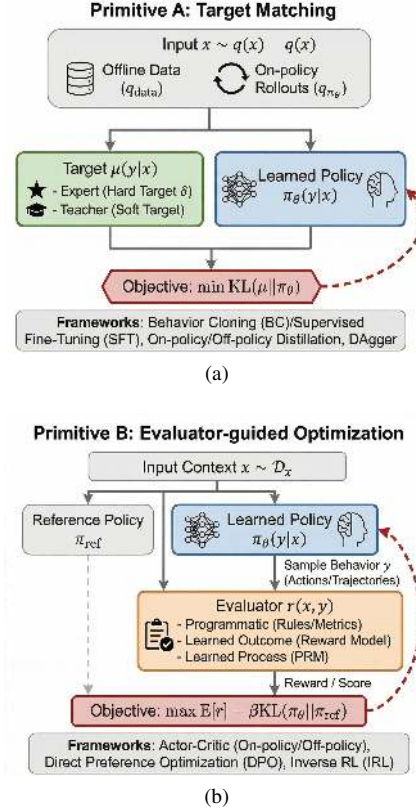


Fig. 3. Overview of the two primary learning frameworks.

B. Primitive B: Evaluator-Guided Optimization

Unlike Primitive A, which directly matches a target distribution, Primitive B optimizes a scalar reward signal. Following the classic Generalized Policy Iteration (GPI) framework [31], we conceptually decouple this primitive into two interacting modules: **Policy Evaluation** and **Policy Improvement**. As shown in Figure 4, policy evaluation (Section II-B1) dictates *how* a generated response or trajectory is scored (i.e., defining the reward $r(x, y)$), whereas policy improvement (Section II-B2) dictates *which algorithm* updates the policy π_{θ} to maximize those scores.

This primitive shares a unified objective: optimizing a conditional policy $\pi_{\theta}(y | x)$ against an evaluator while keeping the policy close to a reference π_{ref} via KL regularization (an especially common practice in LLM alignment):

$$\begin{aligned} \max_{\theta} J(\theta) = & \mathbb{E}_{x \sim \mathcal{D}_x} \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [r_{\phi}(x, y)] \right. \\ & \left. - \beta \text{KL}(\pi_{\theta}(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right]. \end{aligned}$$

Legend: LLM Language domain, AD Driving domain, Signal Reward or Loss mechanism.

B.1 Policy Evaluation (Defining the Reward $r(x, y)$)		
Category	LLM Instantiation	AD Instantiation
B.1.1 Programmatic Fixed, rule-based	response → Verifier (RLVR)	trajectory → collision/comfort metrics
B.1.2 Learned Outcome Parameterized, full seq.	response → LLM-as-a-Judge	trajectory → Trajectory VLM Judge
B.1.3 Learned Process Parameterized, step-wise	step-wise CoT → Process RM (PRM)	step-wise actions → Inverse RL
B.2 Policy Improvement (Optimizing π_θ)		
Category	LLM Instantiation	AD Instantiation
B.2.1 On-policy Actor-critic via rollouts	Optimize via PPO / GRPO	Optimize via PPO / MARL
B.2.2 Off-policy Actor-critic via replay	Rare but emerging (e.g., BAPO [32])	Optimize via SAC / TD3
B.2.3 Direct Pref. Reward implicit in policy	chosen > rejected → DPO loss	safe > unsafe traj. → DriveDPO / TPO loss

Fig. 4. Primitive B (Evaluator-guided optimization) broken down by its two modular choices: the Evaluator (how behavior is scored, top) and the Optimizer (how the policy is updated, bottom). The mappings highlight the analogous concepts across Language Models (LLMs) and Autonomous Driving (AD).

1) **B.1 Policy Evaluation:** Policy evaluation spans two axes: (1) **Reward granularity** (an *outcome* reward over the full trajectory y , or a *process* reward over steps); and (2) **Reward parameterization** (fixed programmatic rules versus learned neural models).

a) **B.1.1 Programmatic Evaluators (Fixed, Non-Parameterized):** Here, $r(x, y)$ is a deterministic rule or metric. In LLMs, this underpins Reinforcement Learning with Verifiable Reward (RLVR), exemplified by DeepSeek-R1’s use of GRPO with automatic correctness signals [33]. In autonomous driving, heuristic reward engineering balances safety, efficiency, and comfort [34]–[42] in difference scenarios across lane-changing and decision under signalized or unsignalized intersections, and for both ego-driving and traffic control. For VLM/VLA driving planners, recent works leverage programmatic rewards with GRPO to enforce formatting, spatial reasoning, and dynamic feasibility (AlphaDrive [43], AutoDrive-R² [17], AutoVLA [44]).

b) **B.1.2 Learned Outcome Evaluators (Parameterized):** Here, a learned model $r_\phi(x, y)$ scores the whole output. In LLMs, this is the foundation of Reinforcement Learning from Human Feedback (RLHF)/ Reinforcement Learning from AI Feedback RLAI, where preference-based reward models are trained via a Bradley–Terry model from pairwise comparisons $\mathcal{D}_{\text{pref}} =$

$\{(x, y^+, y^-)\}$ collected from human rankings or AI feedback (InstructGPT [7], Constitutional AI [45]).

LLMs are also used directly as the reward model $r_\phi(x, y)$. For instance, LLM-as-a-judge paradigm leverages strong LLMs to approximate human preferences (MT-Bench [46]), and LLM-as-a-rubric-judge uses LLMs to score based on custom criteria or rubrics (G-eval [47]).

In driving, this translates to structured, task-specific evaluators: prompting judges to score VLM outputs without human labels (DriveRX [48]), generating reward functions via LLM [49], or using pretrained VLMs to generate semantic rewards from image observations (VLM-RL [50]).

c) **B.1.3 Learned Process Evaluators (Parameterized, Dense):** Here, the evaluator is a learned model that provides process feedback with denser credit assignment than outcome-only reward, e.g., $r_\phi(x, y_{1:t})$ for reasoning steps or a per-time-step reward in an MDP. For LLMs, Process Reward Models (PRMs) validate intermediate reasoning steps (Let’s Verify Step by Step [51], R-PRM [52], OR-PRM [53]). In driving, non-LLM learned dense rewards are often derived via Inverse Reinforcement Learning (IRL), which infers step-wise costs or discriminator-derived rewards from human demonstrations to capture interaction-aware, scenario-specific behaviors [54]–[60]. Recent causal formulations further refine this direction by characterizing when imitation is identifiable under confounding or mismatched sensing, and by providing robust objectives that recover expert-level policies without assuming fully observed causal structure [61], [62].

2) **B.2 Policy Improvement:** Once the reward is given, policy improvement algorithms differ in how they approximate the objective. Here we use the KL-regularized objective as a unifying view: for each input x , we improve $\pi_\theta(\cdot | x)$ against an evaluator while staying close to a reference policy π_{ref} :

$$\max_{\pi(\cdot|x)} \mathbb{E}_{y \sim \pi(\cdot|x)} [r(x, y)] - \beta \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

This objective is shared across LLM post-training (where π_{ref} is the pre-trained / SFT model) and classical RL (where π_{ref} can be a behavior prior or random policy). When π_{ref} is uniform over actions, $\text{KL}(\pi \| \pi_{\text{ref}}) = -H(\pi) + \text{const}$, the KL regularizer is equivalent to an entropy bonus (up to an additive constant).

a) **B.2.1 On-Policy (Actor-Critic via Rollouts):** This family directly optimizes expected reward using on-policy rollouts and a value function (critic) to reduce variance. The canonical PPO clipped surrogate objective

is:

$$\max_{\theta} \mathbb{E}_t [\min (r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)],$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}.$

In the LLM domain, variants stabilize training via asymmetric clipping (clip-higher) (DAPO [63]), sequence-level clipping (GSPO [64]), clipping the importance sampling weight rather than clipping the token updates (CISPO [65]), or substituting the clipping with a more principled constraint based on a direct estimate of policy divergence (DPPO [66]). In driving, PPO and multi-agent variants (MAPPO) are widely applied to mixed-traffic negotiation (e.g., highway merging), cooperative CAV decision-making, and path tracking [67]–[70]. While most papers default to GRPO in driving VLA, more recent papers also explore the application of LLM RL algorithms. WAM-Diff [71] leverages GSPO for a masked-diffusion VLA to optimize sequence-level driving rewards.

b) B.2.2 Off-Policy (Actor-Critic via Replay).: Off-policy methods utilize replay buffers to reuse past experiences for sample efficiency. Due to the low-dimensional, continuous action spaces in classical AD (e.g., steering, acceleration), algorithms like Soft Actor-Critic (SAC) [72] and TD3 [73] are ubiquitous baselines. A representative objective (taken from maximum-entropy actor-critic):

$$\max_{\theta} \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_{\theta}(\cdot|s)} [Q_{\psi}(s, a) + \alpha H(\pi_{\theta}(\cdot|s))],$$

where $\mathcal{B}$ is the replay buffer and Q_{ψ} is learned from off-policy data.

This approach has been rarer in LLMs, though there are efforts and motivations: Gradient accumulation leverages mini-batches to reduce memory peak, which would introduce some amounts of staleness, therefore off-policy updates. BAPO [32] introduces a balanced policy optimization to stabilize off-policy RL for LLM.

In contrast, off-policy RL extensively powers driving for different tasks, ranging from interpretability [74], safe exploration (TD3-ASA [75]), and under different scenarios, such as highway [76] and urban environment with stochastic surrounding traffic [77]. Some recent pipelines also integrate VLM guidance into SAC training loops (VLM-RL [50], DriveMind [78]).

c) B.2.3 Direct Preference Optimization (DPO): DPO bypasses the explicit RL loop by directly optimizing the policy from preference comparisons.

DPO starts from the fact that the KL-regularized objective has a well-known closed-form optimal policy:

$$\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(r(x, y)/\beta).$$

Rearranging gives an equivalent representation of the reward (up to an x -dependent constant that cancels in differences):

$$r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \text{const}(x).$$

Assuming pairwise preferences follow a Bradley–Terry model, DPO replaces π^* with a parameterized policy π_{θ} and maximizes the induced preference likelihood:

$$\max_{\theta} \mathbb{E}_{(x, y^+, y^-) \sim \mathcal{D}_{\text{pref}}} [\log \sigma(r_{\theta}(x, y^+) - r_{\theta}(x, y^-))],$$

where $r_{\theta}(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)}.$

Recent AD research has rapidly adopted DPO to inject human-like driving preferences, rule compliance, and safety bounds into language and vision-based models. Applications range from trajectory-level optimization and predicting human interventions to refining VLA policies via closed-loop takeovers (DriveDPO [79], PrefDrive [80], PPL [81], Multi-PrefDrive [82], CoReVLA [83], LLaViDA [84]), as well as preference-based SAC for specific traffic scenarios [85].

III. ONLINE

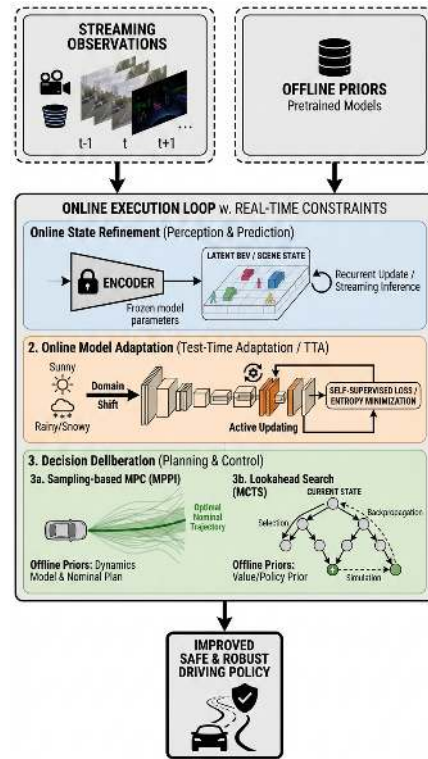


Fig. 5. Overview of online improvement mechanisms.

In Bertsekas’ offline–online perspective [1], the *on-line* loop is the improvement operator at deployment-time. It refines an offline-learned prior using bounded computation under streaming observations and real-time constraints. Such online improvement can occur at multiple layers of the AV stack, i.e., from sensing and scene understanding to planning and control, and these layers differ in what is updated at runtime.

We organize this section into three layers:

- 1) Perception and prediction methods focusing on the streaming nature of observations at deployment-time. We mainly reviewed systems refining the latent or belief state as new frames arrive (§III-A).
- 2) Perception methods that update the model itself to remain robust under domain shift, typically referred as test-time adaptation (§III-B).
- 3) Planning and control methods explicitly optimizing decisions via deliberation. Specifically, we highlight MPPI as a representative of *sampling-based MPC* (§III-C), and MCTS as a representative of *budgeted lookahead search* (§III-D).

We emphasize that this section is *not* an exhaustive survey of planning or control algorithms. Many important alternatives exist (e.g., QP/LTV MPC, NMPC with SQP/RTI, iLQR/DDP, lattice planners, graph search, and game-theoretic or POMDP-style methods), which are reviewed in dedicated planning surveys [3]–[5]. Our goal here is narrower: to clarify how representative online mechanisms consume offline priors (policies, evaluators, and world models) to produce safer and more reliable behavior in mixed traffic.

A. Online State Refinement under Streaming Observations (Sensing & Prediction)

Instead of altering model weights, streaming systems update a latent belief state as new frames arrive. In perception, BEVFormer, and BEVFormer V2 [86], [87] iteratively refine a bird’s-eye-view (BEV) prior using streaming multi-camera features to improve long-horizon temporal consistency. In prediction, recurrent inference architectures carry past trajectory forecasts as priors to progressively refine multimodal predictions in real time, enabling coherent forecasts for newly observed objects [88], [89].

B. Online Model Adaptation under Domain Shift (Perception)

Test-time adaptation (TTA) treats deployment as an unlabeled data stream, updating model parameters at runtime to mitigate shifts under, e.g., weather, illumination, or sensor fidelity, while avoiding catastrophic forgetting. In autonomous driving, TTA methods adapt via entropy minimization (Tent [90]), stochastic weight restoration

for non-stationary streams (CoTTA [91]), time-aware reweighting for correlated streams (RoTTA [92]), and augmenting source data toward target domains (TTA-DAME [93]).

Task wise, TTA is heavily applied to semantic segmentation under gradual or sudden environmental shifts, such as weather changes [94], [95]. It is also utilized for 3D and lane detection, where out-of-distribution (OOD) shifts are mitigated by updating batch normalization parameters on-device [96], leveraging noise-guard regularization (MonoTTA [97]), or efficiently merging checkpoint codebooks (CodeMerge [98]).

C. Online Trajectory Optimization via Sampling-Based MPC (MPPI)

Derived from path-integral optimal control, Model Predictive Path Integral (MPPI) refines continuous trajectories by sampling noisy control sequences, rolling them out through the dynamics, and updating the nominal sequence via an exponentially cost-weighted average. It has been shown that such sampling-based controller can handle complex nonlinearities especially for aggressive autonomous driving where gradient-based planners derail in this regime ([99], [100]).

A closer look into MPPI procedure as described above reveals the interfaces where priors produced by offline learning can be integrated: (1) One direction is smarter proposal distribution as a warm start (TransformerMPPI [101]). (2) A smarter dynamics model that could predict the interaction with surrounding agents, therefore relaxing the assumption that predicted trajectories are given as fixed. For instance, [102] couples MPPI with a neural interaction models that predict surrounding agents’ reactions during rollouts. (3) The constraints at deployment may differ from the learned policy, therefore Residual-MPPI [103] refines the learned policy online for new requirements. (4) By revising the cost functions, different behaviors can be realized. PA-MPPI [104] biases the trajectory into unknown regions for exploration by adding a perception cost.

Conversely, MPPI can be used offline to generate physically feasible demonstrations to enrich training datasets [105].

Recent works integrate VLMs/LLMs with MPPI across two paradigms: **Hierarchical Planning:** LLMs select discrete, high-level navigation primitives while MPPI executes low-level local control (OrionNav [106], [107]). **Semantic Cost Translation:** VLMs translate natural language instructions into semantic cost maps or diffusion-generated nominal trajectories, which MPPI then tracks and refines under dynamic constraints (Words-to-Wheels [108], [109]).

D. Online Lookahead Search for Decision-Making (MCTS)

Monte Carlo Tree Search (MCTS) builds a lookahead tree under a certain simulation budget to balance exploration and exploitation via four phases: Selection, Expansion, Simulation, and Backpropagation. In AD, MCTS typically couples planning with prediction by rolling out surrounding vehicles' behaviors conditioned on ego trajectories.

At the behavior/maneuver level, MCTS enhances neural policy stability in complex interactions, such as highway driving and cooperative multi-agent negotiation [110]–[114], often leveraging inverse planning to recognize other vehicles' goals (deep-MCTS [115], [116]). At the trajectory level, MCTS is adapted for continuous domains using kernel regression, parallelization, and learned nonlinear vehicle dynamics to improve high-speed tracking and safety-critical routing [117]–[121].

There are also AD papers injecting LLM reasoning to guide MCTS value estimation. LLMCTS [122] reports gain from such approach on an unprotected left-turn intersection scenario.

A critical open challenge remains here. For vanilla MCTS, it usually requires a large number of simulations to have a low variance of estimations, which has great difficulties as the depth of search tree grows, so it is hard when the simulation budget is limited. Yet few have adapted advanced techniques such that the performance would not degrade significantly with as small as 5 simulations. Such techniques during tree search could help maintain strong performance even with severely restricted rollout budgets [123], [124].

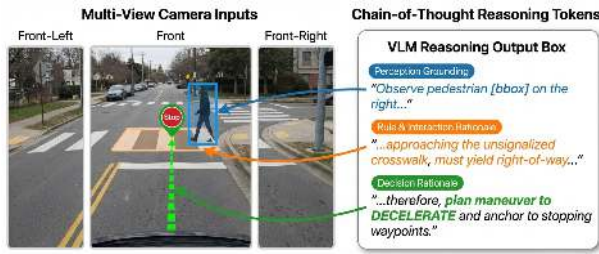


Fig. 6. A grounding example of reasoning tokens in autonomous driving. Given multi-view camera inputs, the model generates an explicit rationale before acting. The token sequence demonstrates distinct **semantics** and **anchoring**: **perception grounding** anchored to an *agent*, **rule justification and interaction rationale** anchored to the *scene*, and finally a **decision rationale** anchored to the planned *maneuver* and *waypoints*. This *chain* structure aligns the offline-learned reasoning with the online behavior execution.

IV. REASONING

In physical AI, “reasoning” broadly refers to inference-time computation expressed through reason-

ing traces, i.e., artifacts such as rationales, decompositions, self-checks, or tool calls generated as intermediate tokens under fixed model weights [163]. Allocating more inference-time compute to produce these traces empirically boosts zero-shot performance [164]–[166]. Theoretically, this mechanism allows constant-size transformers to solve complex problems requiring size- T boolean circuits by generating an $O(T)$ -length reasoning trace [167], highlighting how “more tokens” functions as serial computation.

To ground the concept of reasoning in autonomous driving, Figure 6 illustrates what these intermediate “reasoning tokens” would look like in practice. Rather than directly mapping high-dimensional sensor inputs (e.g., multi-view cameras) to low-level control, VLM/VLA models generate a structured sequence of tokens before outputting a driving command.

From the offline–online perspective (Section II–Section III), reasoning is a cross-cutting capability: the capacity to reason is cultivated offline via curated supervision, while the benefit is realized online by allocating additional computation during execution. Accordingly, we organize this section into two parts: (i) *reasoning datasets* (Section IV-A), which define *what* intermediate tokens should encode; and (ii) *reasoning methods* (Section IV-B), which dictate *how* these tokens are generated and consumed at inference time.

A. Reasoning datasets for driving-relevant capabilities

Reasoning datasets play a different role from standard imitation datasets. Rather than only providing the “correct” action or trajectory, they supervise *intermediate tokens* (e.g., constraints, risk assessments, candidate plans) intended to improve downstream driving decisions. High-quality reasoning supervision in autonomous driving is typically *decision-grounded*: the reasoning trace is tightly coupled to the ego action, references only factors observable within the history window, and avoids generic narratives that do not constrain behavior.

To categorize these datasets systematically, we classify them across four “fit-for-purpose” dimensions (Table I): **Structure** (e.g., free-form, template, chain, programmatic), **Semantics** (what the trace encodes), **Anchoring** (the spatial or functional entity the trace grounds to), and whether they are strongly **Behavior-tied**.

The evolution of these datasets generally follows a semantic progression from scene description to control-tight rationales. Early foundational datasets focus on **Perception Grounding** (**PERC**), teaching the model “what exists” and “where it is.” Datasets in this cluster [126], [134]–[136], [139], [141] frequently utilize free-form or programmatic structures (**FF**, **PROG**) to answer VQA-style queries. Anchored to the overall scene or specific agents (**SCE**, **AGT**), they are only partially or

TABLE I
SUMMARY OF REASONING DATASETS.

Fit-for-purpose:

Structure **FF** free-form, **TPL** template, **MT** multi-turn, **CH** chain, **PROG** programmatic;

Semantics **DEC** decision rationale, **INT** interaction rationale, **RULE** rule justification, **PERC** perception grounding;

Anchoring **SCE** scene, **MNV** maneuver, **WPT** waypoint, **AGT** agent;

Behavior-tied **Y** yes, **P** partial, **N** no.

Cost-to-use:

Supervision **H** human, **M** model (specify), **R** rule/pipeline, **Mix** mixed;

Source Relabeled(base) / New(collection) / Synthetic(sim).

Markers:

* **CAU** explicit causal rationale.

Dataset		Fit-for-purpose				Cost-to-use	
Name	Year	Structure	Semantics	Anchoring	Behavior-tied	Supervision	Source
ARI-CoC [125] * CAU	2025	CH	DEC	MNV	Y	H / M / R (model: GPT-5)	New (collection; internal)
CODA-LM [126]	2024	TPL	PERC	SCE	P	H / M (model: GPT-4V)	Relabeled (CODA [127])
CoVLA [128]	2025	FF	PERC	WPT	Y	R / M (model: VideoLLaMA2-7B)	New (collection; Tokyo)
Drive-R1 [129]	2025	TPL	DEC	WPT	Y	H / M model: GPT-4o	Relabeled (nuScenes)
DriveAction [130]	2025	TPL	DEC	MNV	Y	Mix	New (collection; real-world)
DriveLM [13]	2024	CH	INT	SCE	P	Mix (model: GPT-3.5)	Relabeled (nuScenes) + Synthetic (CARLA)
DriveLMM-o1 [131]	2025	CH	DEC	AGT	P	Mix (model: GPT-4o)	Relabeled (nuScenes)
DriveMLM [132]	2023	TPL	DEC	MNV	Y	Mix (model: GPT-3.5)	Synthetic (CARLA)
SUP-AD (DriveVLM) [14]	2024	CH	INT	MNV	Y	H / R	New (collection; in-house)
Impromptu VLA [133]	2025	CH	DEC	WPT	Y	Mix (model: Qwen2.5-VL 72B)	Relabeled (multi-base) ^a
NuInteract [134]	2025	TPL	PERC	AGT	P	Mix (model: Gemini-Pro)	Relabeled (nuScenes)
NuPrompt [135]	2025	FF	PERC	AGT	P	Mix (model: GPT-3.5)	Relabeled (nuScenes)
NuScenes-QA [136]	2024	PROG	PERC	SCE	N	H / R	Relabeled (nuScenes)
NuInstruct [137]	2024	TPL	INT	AGT	P	Mix (model: GPT-4)	Relabeled (nuScenes)
Reason2Drive [138]	2024	CH	DEC	AGT	P	Mix (model: GPT-4)	Relabeled (nuScenes/Waymo/ONCE)
Talk2BEV-Bench [139]	2023	FF	PERC	AGT	P	H	Relabeled (nuScenes)
DriveGPT4 (Inst.) [140]	2024	MT	DEC	MNV	Y	Mix (model: ChatGPT)	Relabeled (BDD-X)
NuScenes-MQA [141]	2024	PROG	PERC	SCE	N	H / R / M (model: GPT-4)	Relabeled (nuScenes)
WOMD-Reasoning [142]	2024	PROG	INT	WPT	P	R / M (model: ChatGPT)	Relabeled (WOMD)
DRAMA [143]	2023	TPL	DEC	AGT	P	H	New (collection; Honda/Tokyo)
Rank2Tell [144]	2024	TPL	DEC	AGT	P	H	New (collection; Honda)
NuScenes-SpatialQA [145]	2025	PROG	PERC	AGT	N	R	Relabeled (nuScenes)
STSBench [146]	2025	PROG	INT	SCE	P	H / R	Relabeled (nuScenes)
SURDS [147]	2024	PROG	PERC	AGT	N	R	Relabeled (nuScenes)
VLADBench [15]	2025	TPL	DEC	SCE	P	H / R	Relabeled (multi-base)
DrivingContexts (ContextVLM) [148]	2024	TPL	PERC	SCE	N	Mix	New (collection; mixed sources)
DriveCoT [149]	2024	CH	DEC	MNV	Y	R	Synthetic (CARLA)
LMDrive [12]	2024	TPL	DEC	SCE	Y	R / M (model: ChatGPT)	Synthetic (CARLA)
LingoQA [150]	2024	FF	DEC	SCE	P	H / M (model: GPT-3.5)	New (collection; real-world)
OmniDrive [151]	2024	PROG	DEC	WPT	Y	H / M / R (model: GPT-4)	Relabeled (nuScenes)
DriveX (Senna) [152]	2024	PROG	DEC	WPT	Y	R / M (model: LLaVA-v1.6-34B)	New (collection; not publicly released)
nuScenesR ² -6K [17]	2025	CH	DEC	WPT	Y	M	Relabeled (nuScenes)
DRAMA-ROLISP [153]	2023	TPL	DEC	AGT	P	H / M (model: GPT-4)	Relabeled (DRAMA)
BDD-X [154]	2018	TPL	DEC	MNV	Y	H	Relabeled (BDD)

^a Base datasets: Mapillary [155], ONCE [156], NAVSIM [157], nuScenes [158], Waymo [159], ArgoverseV2 [160], KITTI [161], IDD [162].

not at all tied to behavior (**P**, **N**). This disconnect has led to concerns regarding whether the visual reasoning

is genuinely causal for driving [129].

Moving closer to execution, a second cluster targets **Interaction Rationale** (**INT**) and rule justification. Works such as [13], [14], [137], [142] explicitly encode “who should yield” and “where conflicts arise,” adopting template or chain structures (**TPL**, **CH**) to break down complex social interactions anchored to specific agents or upcoming maneuvers.

Finally, recent efforts prioritize **Decision Rationales** (**DEC**) that are strictly decision-grounded [125], [129], [130], [133], [138], [140]. These datasets directly align tokens with control outputs, explicitly anchoring thoughts to ego maneuvers (**MNV**) or specific waypoints (**WPT**), ensuring they are strongly tied to behavior (**Y**).

Within this category, [125] sets a high standard for reasoning in practice. They argue that reasoning data must be strictly correlated with the ego trajectory to avoid the superficial narratives and causal confusion prevalent in earlier datasets. To address these gaps, they construct a Chain of Causation (CoC) dataset by: (1) defining a closed-set of high-level driving decisions that directly correspond to low-level ego trajectories; (2) associating each reasoning trace with a direct driving decision, ensuring it includes *only* the causal factors that motivate that specific decision; and (3) selecting decision-grounded keyframes to guarantee all causal factors can be observed within the history window. This rigorous approach provides clear supervision for learning true decision causality.

B. Reasoning methods: generating and consuming intermediate tokens

We now summarize inference-time methods for reasoning. We categorize these methods by two orthogonal choices regarding the intermediate tokens: their **source** (internal model-generated vs. external augmentation) and their **consumer** (internal self-improvement vs. external interaction with tools). This yields four distinct method families.

1) *Model-generated reasoning traces (internal provenance)*: This family focuses on prompting strategies that shape the model’s internal reasoning traces. Foundational techniques like **Chain-of-Thought (CoT)** [165], [166] elicit step-by-step rationales to improve multi-step problem solving. This concept has been widely expanded into methodologies that **decompose** complex queries into simpler, sequential sub-tasks [168]–[172], or leveraging **extra structures** like tree, or graph to combine intermediate thoughts into searchable trees and graphs, such as Tree-of-Thoughts (ToT) [173], and Graph-of-Thoughts (GoT) [174].

As for the applications to the autonomous driving domain, the CoT and decomposition are used heavily. One typical application is to improve multimodal perception

and provide explicit explainability (Dolphins [175], CoT-Drive [176], X-Driver [177]). The reasoning traces are also used to evaluate the risks. Think-Driver [178] uses CoT to evaluate the risks of current driving maneuvers to support rational driving decisions, [179] also uses CoT for uncertainty-aware object interaction tasks to support interpretable decision making. CoT4AD [180] explicitly models a perception-question-prediction-action CoT across multiple driving tasks.

2) *Externally augmented reasoning traces (external provenance)*: Rather than relying solely on model weights, external augmentation injects dynamic tokens, such as retrieved passages, stored experiences, or structured graphs, into the reasoning trace at inference time to reduce hallucination and bring knowledge without retraining. For this category, we can split it roughly to be (1) **retrieval** and (2) **memory**.

Retrieval-Augmented Generation (RAG) [181] typically supplies free-form, unstructured text, which couples a retriever with a generator to explicitly condition outputs on external evidence. This concept has evolved into adaptive retrieval like Self-RAG [182], which decides when to retrieve and emit reflection-style signals to regulate evidence use during generation; hierarchical retrieval like RAPTOR [183] for long corpora and retrieval at the appropriate granularity; and structured graph-based retrievers such as StructRAG [184] and GraphRAG [185] for knowledge-intensive reasoning tasks.

Beyond static retrieval, **memory** mechanisms maintain a stateful, often writable store for contents like past interactions, and carry information beyond the context window. The memory can be in the architectural sense ([186], [187], [188], [189]), where mechanisms (e.g., recurrence, caching, compression, memory tokens) are developed to carry information forward over long horizons. In the agent context, **memory** represents a different pipeline: the agents writes new entries during deployment (events, reflections, skills), and later retrieves a small subset back into the context window, as demonstrated by generative agents [190], MemGPT [191], and Voyager [192].

In the autonomous driving domain, **retrieval** is used to ground reasoning by fetching historical cases or regulatory rules. RAG-Driver [193], RAC3 [194], and RAD [195] retrieve similar past driving cases to provide context for handling rare corner situations and improving reliability. Other works fetch safety guidelines from traffic documents or knowledge graphs to enhance trustworthiness, such as Driving with Regulation [196], KnowVal [197], and SafeDriveRAG [198]. The **agentic memory** paradigm is also heavily adopted to handle long-tail or out-of-distribution (OOD) scenarios. Models like Dilu [199], LeapAD [200], LeapVAD [201], Drive Like a Human [202], DriVLMe [203], and Actor-

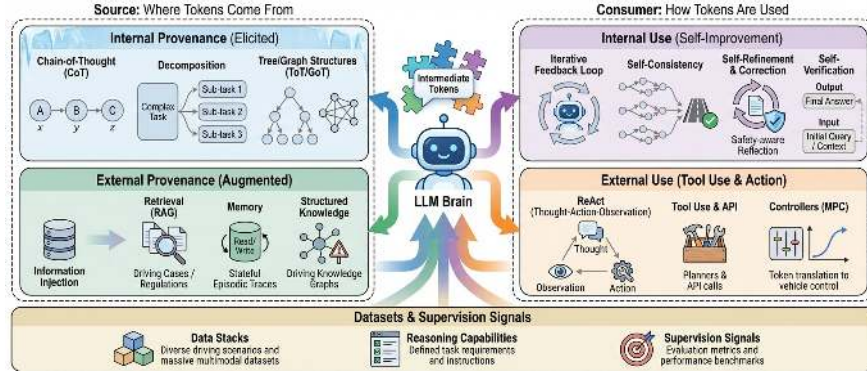


Fig. 7. Taxonomy of inference-time reasoning by token source and token consumer. Intermediate tokens can be produced internally (elicited by prompting and structured reasoning) or injected externally (retrieval, memory, and structured knowledge). These tokens are then consumed either for internal self-improvement (self-consistency, reflection, verification) or for external interaction (tool use, planner/controller interfaces). The bottom panel illustrates what datasets and supervision signals can provide for these reasoning behaviors.

Reasoner [204] include memory components to continuously accumulate experience. Furthermore, MTR-Drive [205] and LiloDriver [206] explicitly leverage this memory to navigate OOD situations.

3) *Reasoning traces for self-improvement (internal use)*: Start from this part, we shift our focus from where reasoning tokens come from (model-generated vs. externally injected), to where they are consumed. Methods in this family use reasoning traces internally to improve the model's own outputs.

For general LLMs, self-improvement strategies include sampling diverse reasoning paths for self-consistency [207], performing backward self-verification to validate conclusions [208], or iteratively self-refining outputs using internal feedback without external supervision [209]. Expanding on these, Natural Program [210] enforces deductive reasoning formats to further enhance the rigor of the self-verification process.

For autonomous driving, these self-improvement mechanisms are critical for robustness and safety. Chain-of-thought with self-consistency (COT-SC) is used by OFF-EMMA [211] to improve off-road trajectory planning accuracy, while [212] applies self-consistency to improve driving QA reliability. Furthermore, intermediate tokens enable explicit safety-aware reflection before executing a final maneuver. ReflectDrive [213] runs a reflection mechanism to identify and correct unsafe trajectory tokens iteratively. CF-VLA [214] proposes an inference-time self-reflective mechanism that uses counterfactual reasoning to revise meta-actions before trajectory generation. Similarly, SVBRD-LLM [215] generates behavioral-rule hypotheses and iteratively refines them to build a high-confidence rule library.

4) *Reasoning traces for tool use and action (external use)*: In this family, intermediate tokens are executable actions (e.g., API calls, code, planner queries) that interact with external environments, closing the reasoning

loop with observational feedback.

In LLM, general approaches like ReAct [216] force the model to alternate between generating natural-language reasoning and explicit action tokens to interface with external tools. MRKL [217] treats the LLM as a router that selects specialized modules, while Program of Thoughts (PoT) [218] expresses the reasoning process as a program executed by an external computer to derive the final answer.

In the AV stack, this external interaction is realized at both macro and micro levels. At the macro orchestrator level, models alternate between text reasoning and tool invocation to interact with the driving stack or simulators. DriveAgent-R1 [219] and Agent-Think [220] integrate text reasoning with active perception and tool evaluation. Agent-Driver [221] and PlanAgent [222] build agent loops around traditional motion planning components, while ChatSim [223] converts natural-language commands into scene-edit actions using simulator feedback. At the micro level, intermediate tokens are consumed directly by low-level controllers. LanguageMPC [224], VLM-MPC [225], and LVLM-MPC [226] leverage language models to output symbolic task commands or explicit parameters (e.g., desired speed, headway) to guide Model Predictive Control (MPC) engines. Similarly, CALMM-Drive [227] generates multiple candidate plans with confidence scores, relying on an external refinement process to find the optimal trajectory.

V. WORLD MODEL

World models learn action-conditioned state evolution, enabling the ability to score counterfactual futures. In the context of autonomous driving, these models primarily serve two roles: (1) Offline (as simulators) to generate closed-loop environments for training, testing, and covering long-tail distributions. (2) Online (as search engines)

to roll out candidate trajectories at decision time, thereby facilitating better, forward-looking decisions.

A. Offline world model

Efforts in offline simulation have increasingly shifted towards large-scale generative modeling. Early multimodal efforts like GAIA-1 [228] demonstrated the ability to capture expectations of future events, which has been followed by models like DriveDreamer [229] that are trained entirely on large real-world driving datasets. [230] leverages Cosmos-Transfer1 with ControlNet to generate realistic driving videos of rare but critical edge cases. Recently, industry-scale models have pushed these boundaries further; for instance, the Waymo World Model [231], built upon Google DeepMind’s Genie 3 [232], leverages an advanced general-purpose architecture capable of simulating exceedingly rare driving events.

Beyond general video generation, another cluster of works focuses on high-fidelity sensor synthesis and closed-loop environment reconstruction. UniSim [233] proposes a neural simulator that converts a single driving log into a realistic multi-sensor simulation. Building on scene reconstruction, SceneDiffuser++ [234] enables city-scale point A-to-B simulation, while RAD [235] leverages 3D Gaussian Splatting (3DGS) to build photorealistic environments that bridge the open-loop gap and improve safety-critical handling. Furthermore, open-source frameworks like Nvidia’s AlpaSim [236] provide robust platforms for closed-loop simulation with accurate sensor data and vehicle dynamics.

To explicitly target the generation of challenging situations and improve model efficiency, specialized scenario generators are also heavily utilized. TrajGen paired with I-Sim [237] captures realistic driver behaviors from human demonstrations to populate the environment. Scengine [238] generates diverse traffic scenarios using operators inspired by biological genetic processes. Meanwhile, SEM2 [239] introduces a semantic-masked recurrent world model that concentrates on driving-relevant features, significantly improving sample efficiency and overall robustness.

B. Online world model

When deployed online, world models act as dynamic rollout engines that integrate tightly with motion planners and control algorithms. A prominent strategy is coupling the world model with sampling-based or predictive controllers. For example, DAWM [240] integrates a driving-aware world model with MPPI control to optimize ego trajectories in complex situations, such as unsignalized T-intersections in CARLA. Similarly,

AdaptiveDriver [241] builds an MPC planner that adaptively unrolls different world models conditioned on the predictions of a separate behavior network. To ensure these models generalize across different platforms and vehicle parameters, VDD (Vehicle Dynamics embedded Dreamer) [242] explicitly separates ego-vehicle dynamics from environmental dynamics.

Another distinct approach is the use of world models for explicit occupancy forecasting and multimodal trajectory search. Drive-WM [243] is designed to be compatible with existing planning models, enabling the search for optimal trajectories by imagining rollouts from distinct driving maneuvers. Extending this to volumetric predictions, Drive-OccWorld [244] forecasts future 4D occupancy conditioned on actions, which is then plugged into a planner utilizing occupancy-based costs to select safe trajectories. OccLlaMA [245] unifies this paradigm by introducing an occupancy-language-action generative model, employing a shared multimodal vocabulary to natively complete driving tasks, including motion planning, within the imagined environment.

VI. CONCLUSIONS AND OPEN QUESTIONS

Viewed through the offline-online two-loop framework, reasoning-based autonomous driving can be understood as learning reusable priors offline and then spending bounded compute online to refine decisions via reasoning rather than purely reactive. In this survey, we used the offline-online framework to organize (i) offline learning methods, (ii) online improvement mechanisms, (iii) the reasoning layer that bridges the two loops by turning rationales, constraints, and structured plans into actionable decision-time computation, (iv) and world models serve as simulators offline and rollout engine online. We further analyze some directions and open questions for future research, which we hope could offer the community novel insights.

A. On-Policy Distillation for Closed-Loop Autonomous Driving

Most learning pipelines for autonomous driving rely on off-policy supervision—either behavioral cloning from logged human demonstrations or offline distillation from fixed datasets. While effective for learning strong priors (as discussed in Section II), these approaches suffer from covariate shift: once deployed, a learned policy induces a different state distribution than the dataset, and small errors can compound over long horizons. A promising future direction is on-policy distillation, but under modern foundation-model and simulator capabilities. Concretely, the deployable student policy is rolled out in closed-loop to generate the states it actually visits, and a stronger “teacher” provides corrective supervision on those states.

On-policy distillation has gained momentum in the LLM community because teacher models are strong, rollouts are cheap, and intermediate reasoning traces enable scalable inspection and filtering. Similar conditions are now emerging for autonomous driving. First, increasingly capable reasoning-based VLA systems can act as high-quality teachers that provide not only action labels but also interpretable rationales (e.g., Alpamayo [125]). Second, the growth of world models and high-fidelity simulators (e.g., UniSim [233], AlpaSim [236]) makes it feasible to safely unroll the student policy to gather the on-policy state distribution.

B. Compute-Efficient Online Planning and Control under Domain Shift

A key asymmetry in today’s literature is that “domain shift” is often treated as a perception problem (Section III-B), while planning and control are implicitly assumed to remain well-posed once the state is accurate. In mixed traffic, however, the online planner faces its own forms of shift, for instance, vehicle dynamics drift (e.g., payload, tire, road friction), and the implicit objectives such as comfort and risk tolerance vary across fleets. When large VLM/VLA components are inserted into the online loop, for instance, to evaluate candidates, or predict interactive responses, the resulting planning operator can become both *compute-heavy* and *shift-fragile*.

This motivates online methods that are robust under domain shift and budget-aware. One direction is to revisit ideas from search-based RL, as discussed in Section III-D, where methods like [123], [124] enable low-variance estimation for MCTS with a magnitude lower number of rollouts. For autonomous driving specific context, the analogue is a decision module that produces a safe, feasible candidate quickly (under small rollout budget), while explicitly tracking uncertainty and triggering conservative fallbacks.

C. Capability-Specific Reasoning Datasets

Recent reasoning datasets for driving and embodied agents have primarily targeted broad, general-purpose capabilities—such as scene description, commonsense justification, or generic trajectory planning. While useful for establishing baseline reasoning behavior, they may not provide strong supervision on the specialized cognitive skills required for safety and legality in real-world autonomous driving. A future direction is the development of *capability-specific* reasoning datasets designed to explicitly cultivate the kinds of reasoning that must occur online.

We are beginning to see early successful efforts in this direction. For instance, to specifically tackle the capability of *causal reasoning* and mitigate the causal confusion

found in general datasets, Alpamayo [125] introduced the Chain of Causation (CoC) dataset. Another gap is the capability for explicit *regulatory and legal-cases reasoning*. While inference-time methods like Driving with Regulation [196] have shown the value of injecting traffic rules into the pipeline, there is a lack of large-scale, capability-specific datasets for regulation or legal cases (related to driving). Such datasets would pair a concrete scenario (e.g., sensor history, map context, agent interactions) with a reasoning trace that explicitly retrieves similar past law cases or applies specific jurisdictional traffic rules. This would train the model not just to generate a safe trajectory, but to reason about *why* an action is legally justified in an edge case, explicitly representing uncertainty when rules conflict or permit multiple options.

Future work should focus on datasets that target these specific reasoning skills, which could be causal inference, case-based legal reasoning, or multi-agent negotiation. By shifting from datasets that teach “general driving ability” to a suite of capability-specific benchmarks, the community can provide rigorous reference standards for what reasoning tokens should contain. This will directly support automated verification modules and ensure that advances in VLA reasoning translate into reliable, auditable autonomy.

D. Closed-Loop Self-Play in Generative World Models

Large-scale self-play in high-throughput simulators has recently demonstrated that complex interactive driving behaviors and strong robustness can emerge from purely synthetic closed-loop experience, achieving state-of-the-art results on multiple realistic benchmarks [40]. As generative world models emerge [228], [231], [243], closing the RL loop entirely inside the world models becomes a potential next step. The core premise is to introduce the complex and realistic dynamics into self-play, which could derive different driving policies than those emerged from simulators.

A central challenge is that today’s driving world models are often most fragile in regimes that matter for robustness: rare events, near-collisions, and off-distribution ego or other-agent behaviors. For instance, [246] highlights that world models trained predominantly on expert and safe trajectories may fail to faithfully track hazardous or non-expert behavior, limiting their utility for reliable policy evaluation and selection. World-model self-play offers a way to close this gap: policy self-play generates diverse interactive rollouts (including near-failure regimes), which become the data for improving the world model; the improved world model then enables more realistic closed-loop training, yielding a curriculum that could learn the rare cases.

ACKNOWLEDGMENT

Generative AI-assisted tools were used in preparing this manuscript. Google Gemini was used to render and adjust the figures. OpenAI GPT-5.2 and Google Gemini were used to edit and enhance the writing. All outputs were reviewed and revised by the authors, who take full responsibility for the content of this article.

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